

## Project Design Document

### Project Title

YUMBYTE

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YUMBYTE  
MAKE IT UNIQUE

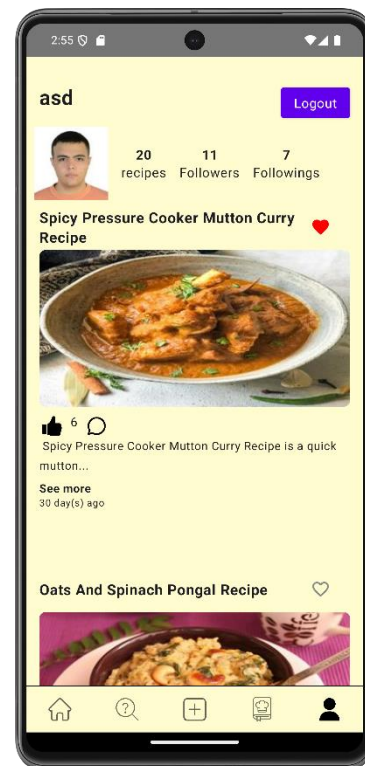
|    |                                          |   |
|----|------------------------------------------|---|
| 12 | <b>Contents</b>                          |   |
| 13 | Introduction.....                        | 3 |
| 14 | Key Features .....                       | 3 |
| 15 | Low-Level Design .....                   | 6 |
| 16 | Technologies .....                       | 6 |
| 17 | <b>A. Programming Languages</b> .....    | 6 |
| 18 | <b>B. Frameworks and Libraries</b> ..... | 7 |
| 19 | <b>C. Tools and Services</b> .....       | 7 |
| 20 | Appendix .....                           | 9 |
| 21 |                                          |   |
| 22 |                                          |   |

## 23 Introduction

- 24 • **Problem Statement:** In today's fast-paced digital world, food enthusiasts face challenges in  
25 discovering new and personalized recipes that align with their tastes and preferences. While there  
26 are numerous recipe-sharing platforms, many lack the social connectivity and personalized  
27 experiences that users crave. Existing platforms often provide static content, making it difficult  
28 for users to discover new recipes that cater to their specific dietary needs and culinary interests.  
29 This disconnect overwhelms users by the sheer volume of content, without a streamlined way to  
30 explore and interact with recipes that truly resonate with them.
- 31 • **Project Objective:** Our project aims to solve these challenges by creating a dynamic platform that  
32 not only allows users to share their culinary creations but also connects them with like-minded  
33 individuals. By integrating a machine learning model, our platform will provide personalized  
34 recipe recommendations tailored to each user's unique preferences, thus enhancing their overall  
35 culinary experience. Inspired by the social nature of Instagram, our platform seeks to build a  
36 vibrant community of food lovers, where users can discover, share, and engage with a diverse  
37 array of recipes in an interactive and personalized manner.

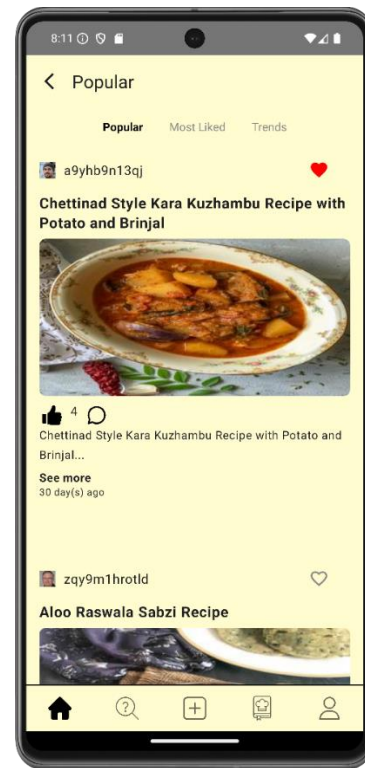
## 38 Key Features

**1- User Accounts:** Users can create and manage their accounts, enabling full participation in the platform. This includes features like account verification, profile customization updating profile pictures, and managing social interactions such as viewing and following other users and viewing followers and followings.

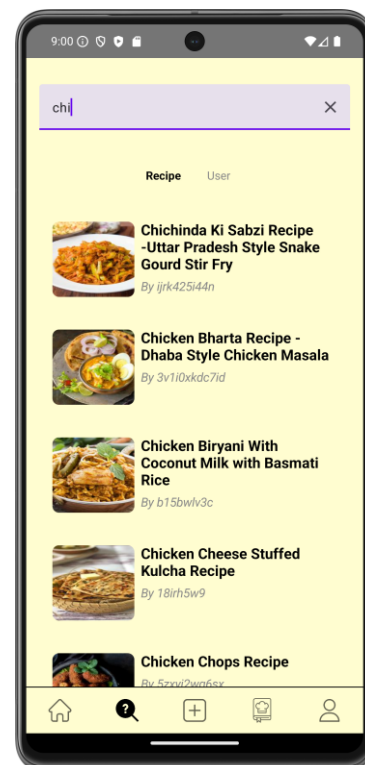


**2- Home Screen Categories:** The home screen is the central hub for exploring various recipe categories:

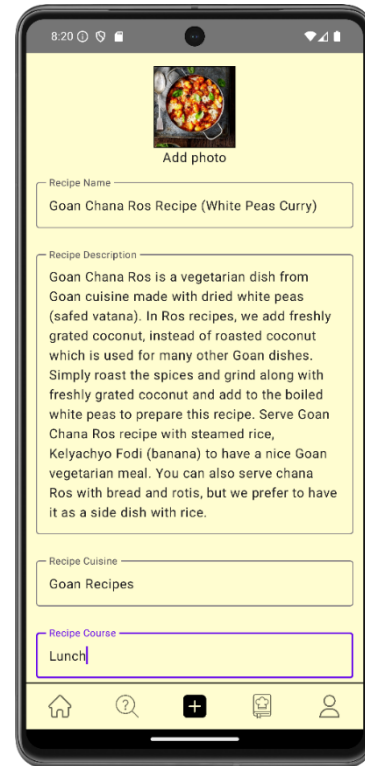
- **Popular Recipes:** Discover recipes that are trending and widely appreciated by the community.
- **Most Liked Recipes:** Explore the top recipes with the most likes from users.
- **Trending Recipes:** Recipes that are currently gaining popularity.
- **Favorites:** Users can save and manage recipes they like in a customizable favorites list, allowing easy access and organization of their preferred recipes.



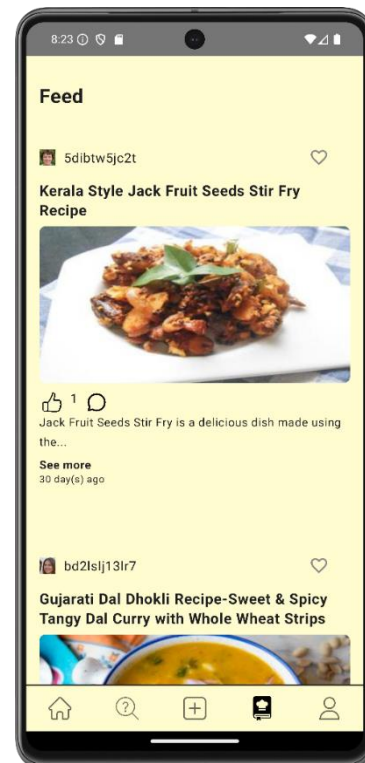
**3- Search Functionality:** The search feature allows users to find specific recipes and other users within the app. It provides a straightforward way for users to discover content that matches their interests and needs, ensuring they can quickly locate recipes or connect with other food enthusiasts.



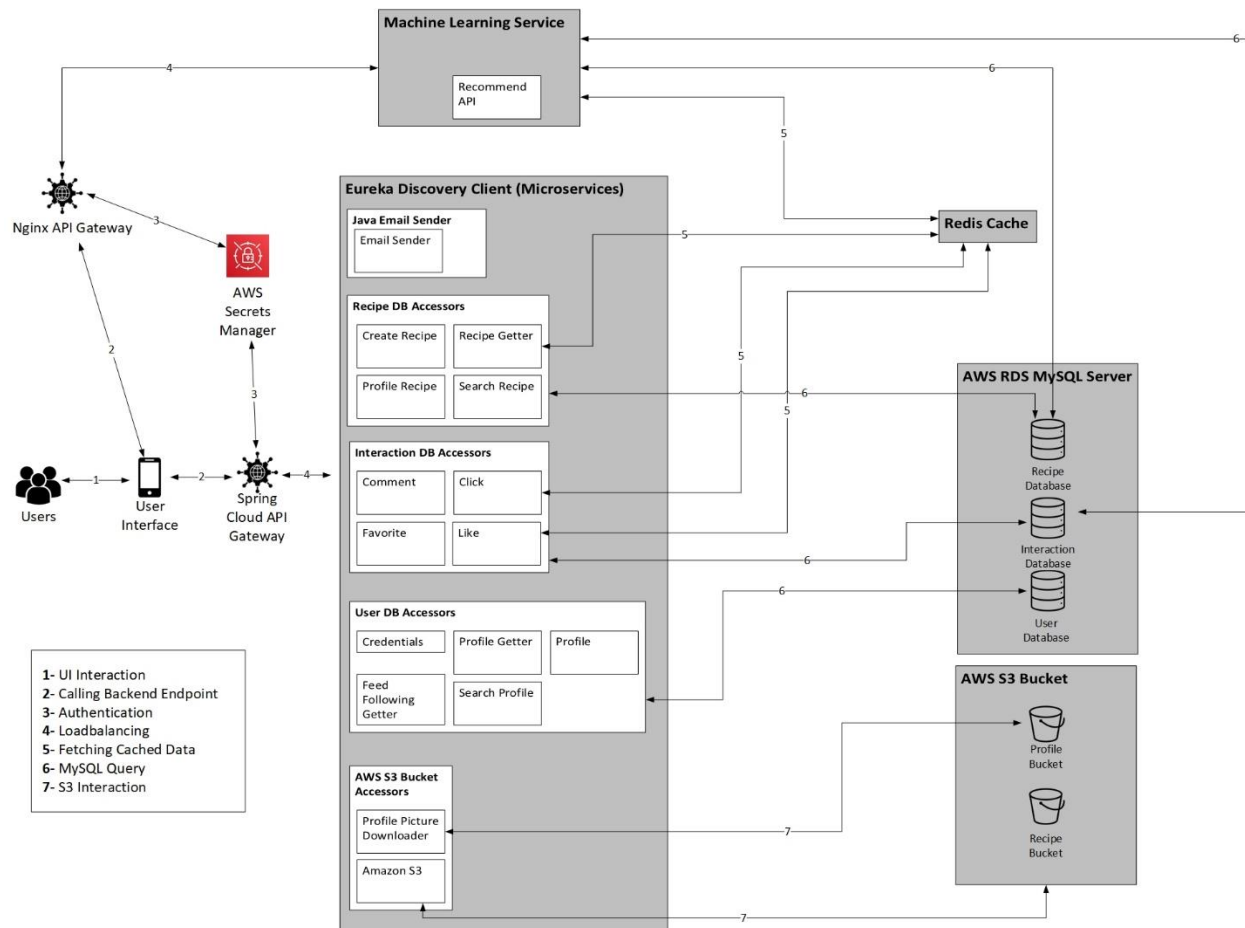
**4- Recipe Creation and Sharing:** Users can create and share their recipes with the community. This feature includes an intuitive interface for entering recipe details such as ingredients, instructions, cooking time, and uploading photos. By sharing their culinary creations, users contribute to the platform's growing collection of recipes, enriching the community with diverse food options.



**5- Personalized Feed:** The feed provides a personalized stream of recipe recommendations, powered by a machine-learning model that analyzes user behavior, likes, and interactions. This ensures that users consistently see content tailored to their tastes, keeping them engaged and inspired to try new dishes.



## Low-Level Design



## Technologies

### A. Programming Languages

- **Java (Spring Boot):** Java is a robust, object-oriented programming language that's widely used for backend development. Spring Boot, a framework based on Java, simplifies the development of web applications by providing pre-configured templates and tools. It's particularly suitable for this project due to its strong support for creating scalable and maintainable web services.
- **Kotlin:** Kotlin is a modern, statically typed programming language that's fully interoperable with Java. It's particularly well-suited for Android development and is used in this project to build a responsive and efficient mobile application. Kotlin's concise syntax and strong tooling support make it a great choice for this project.
- **MySQL:** MySQL is a popular open-source relational database management system. It's used in this project for storing and managing structured data, such as user profiles, recipes, and interactions. MySQL's reliability and strong query capabilities make it ideal for handling the complex data relationships involved in this application.
- **Python (Machine Learning):** Python is a versatile programming language with extensive libraries for data science and machine learning. In this project, Python is used to implement machine-

learning algorithms that power personalized recipe recommendations. Its simplicity and strong community support make Python an excellent choice for this task.

## B. Frameworks and Libraries

- **Spring Boot Cloud (Feign, API Gateway, Naming Server):** Spring Boot Cloud is a suite of tools and frameworks that simplifies the development of distributed systems and microservices. In this project, Feign is used to create declarative web service clients, making it easier to communicate between microservices. The API Gateway acts as a single entry point, routing requests to the appropriate microservices, while the Naming Server provides service discovery, allowing microservices to locate each other dynamically. These tools together enable scalable, maintainable, and efficient communication across different parts of the application.
- **Jetpack Compose:** Jetpack Compose is a modern toolkit for building native Android UIs, streamlining the process of UI development with less code and more powerful tools. In this project, Jetpack Compose is utilized to create a dynamic and responsive user interface, providing an intuitive and smooth user experience by allowing developers to build and manage complex layouts with ease.
- **Flask:** Flask is a lightweight web framework for Python that allows for the quick development of web applications and APIs. In this project, Flask serves as the backend framework for exposing machine learning models via RESTful APIs, enabling seamless integration between the machine learning components and other parts of the system.
- **NumPy:** NumPy is a fundamental library for numerical computing in Python. It provides support for arrays, matrices, and a large number of mathematical functions to operate on these data structures. In this project, NumPy is used for efficient numerical operations and data manipulation, which is essential for processing and analyzing data in the machine-learning components of the system.
- **Scikit-Learn:** Scikit-Learn is a robust machine learning library in Python, offering various tools for data preprocessing and implementing algorithms. It is used in this project for tasks like data transformation (using 'OneHotEncoder' and 'MinMaxScaler'), text processing ('TfidfVectorizer'), and implementing the K-Nearest Neighbors (KNN) algorithm, which powers personalized recipe recommendations.
- **MySQL Connector:** MySQL Connector is a Python library that facilitates interaction with MySQL databases. In this project, it connects the Python-based machine learning module to the MySQL database, enabling efficient data retrieval and storage, which is crucial for implementing personalized recommendations.
- **Coil:** Coil (Coroutine Image Loader) is an image loading library for Android optimized with Kotlin Coroutines. It is used to efficiently load and display images within the app, ensuring that visual content is managed seamlessly and without performance issues. In this project, Coil is configured to use 25% of the available RAM and 50 MB of disk space for caching images, which helps in speeding up image loading and reducing network requests.

## C. Tools and Services

- **Redis:** Redis is an in-memory data structure store used as a database, cache, and message broker. In this project, Redis is utilized to cache frequently accessed data such as the number of likes and clicks for each recipe, as well as storing some recipe IDs for quick access. This caching approach significantly reduces the load on the MySQL database and enhances the system's overall performance, ensuring faster data retrieval and a more responsive user experience.

- 101 • **Retrofit:** Retrofit is a type-safe HTTP client for Android and Java, used for making API requests and  
102 handling responses. In this project, Retrofit facilitates communication between the Android  
103 mobile application and backend services, ensuring that data is exchanged smoothly and  
104 efficiently.
- 105 • **Kotlin Coroutines:** Kotlin Coroutines are used for managing asynchronous tasks in a more  
106 readable and efficient manner. In this project, coroutines are employed to handle background  
107 operations like network requests and database interactions without blocking the main thread,  
108 enhancing the app's responsiveness.
- 109 • **AWS S3:** AWS S3 (Simple Storage Service) is a scalable object storage service used to store and  
110 retrieve any amount of data at any time. In this project, AWS S3 is used to store and manage  
111 images associated with recipes and users, ensuring scalable and reliable storage solutions.
- 112 • **AWS RDS:** AWS RDS (Relational Database Service) is a managed relational database service that  
113 provides scalable database instances. In this project, AWS RDS is used to host the MySQL  
114 database, providing a managed environment with automated backups, patching, and scaling,  
115 which enhances database performance and reliability.
- 116 • **AWS Secrets Manager:** AWS Secrets Manager is a service for securely managing and accessing  
117 sensitive information, such as API keys. In this project, AWS Secrets Manager is used to store and  
118 manage the API key required for authenticating API requests.



## 119 Appendix

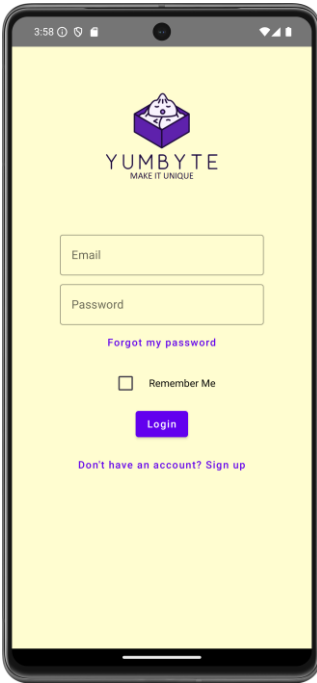


Figure 1. Login Page

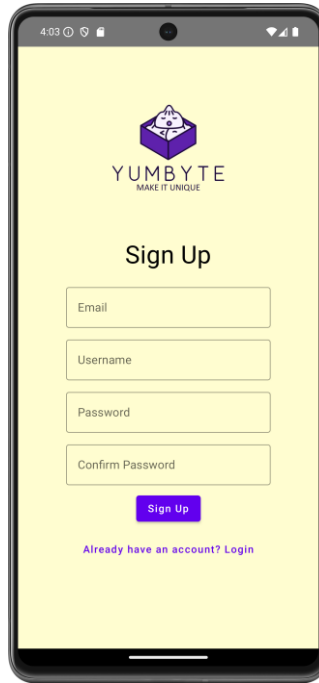


Figure 2. Signup Page

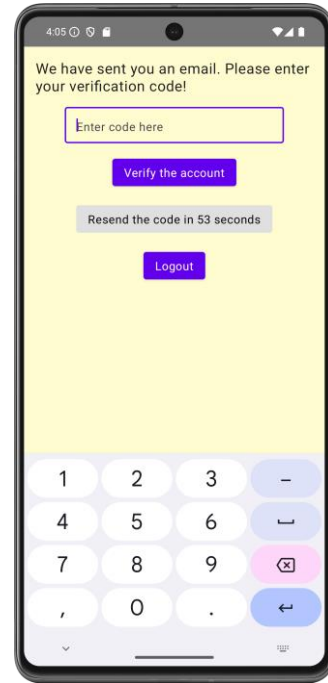


Figure 3. Verification Page

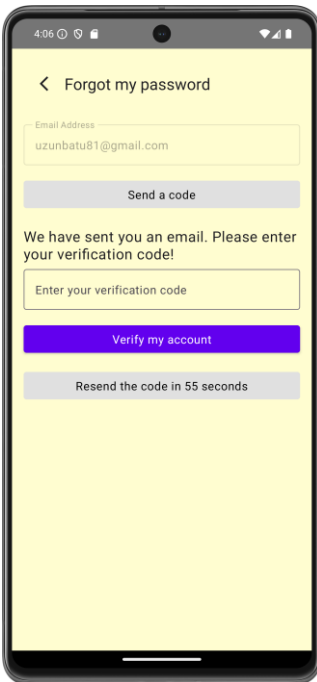


Figure 4. Change Password Page

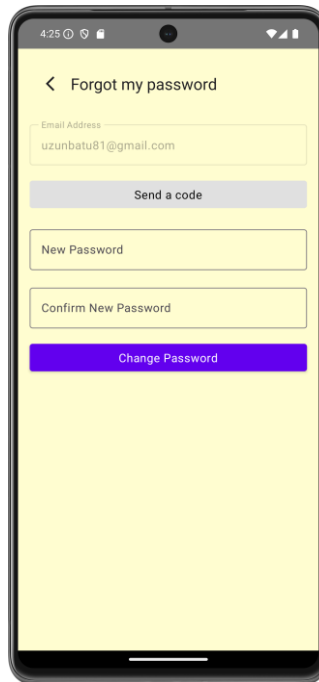


Figure 5. Change Password Page



Figure 6. Home Page



Figure 7. Recipe Details Page



Figure 8. Recipe Details Page

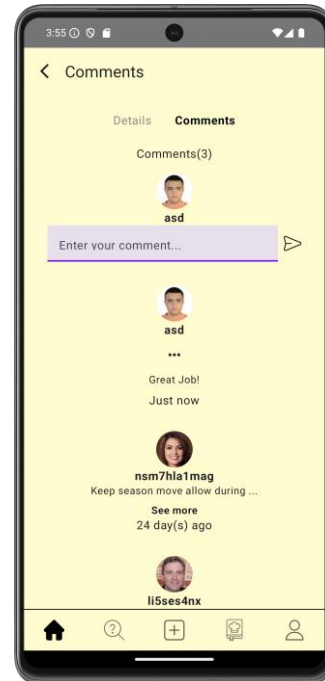


Figure 9. Comments Page



Figure 10. Profile Page

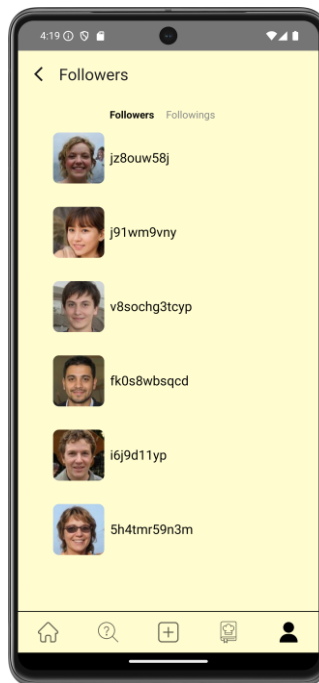


Figure 11. View Followers Page



Figure 12. View Followings Page

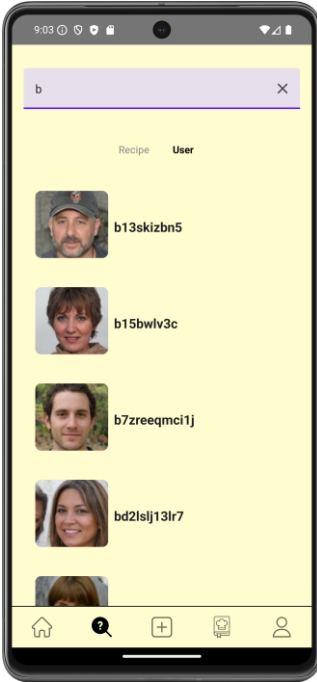


Figure 13. Search User Page